Hellenic Complex Systems Laboratory

Relation of Positive and Negative Predictive Values of Diagnostic Tests

Technical Report XII

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2015

Relation of Positive and Negative Predictive Values of Diagnostic Tests

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Short Description of the Demonstration

This Demonstration examines the relation of the negative predictive value (NPV) and the positive predictive value (PPV) of a diagnostic test for normally distributed nondiseased and diseased populations. Different levels of prevalence of the disease are considered. The mean and standard deviation of the populations, measured in arbitrary units, are used.

Search Terms

sensitivity, specificity, diagnostic test, clinical accuracy, diagnostic accuracy, normal distribution, positive predictive value, negative predictive value

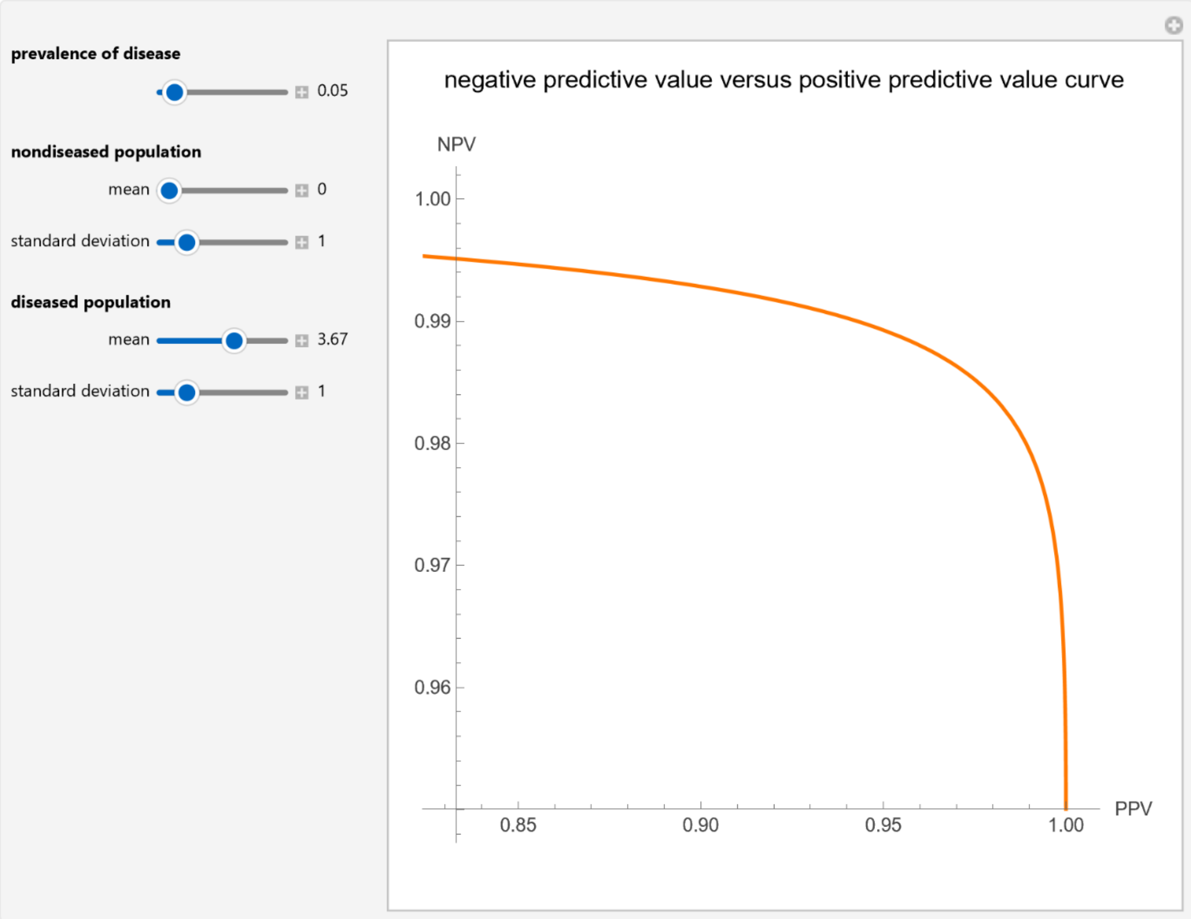

Figure 1: *Negative predictive value versus positive predictive value curve plot of a diagnostic test, with the settings shown on the left.*

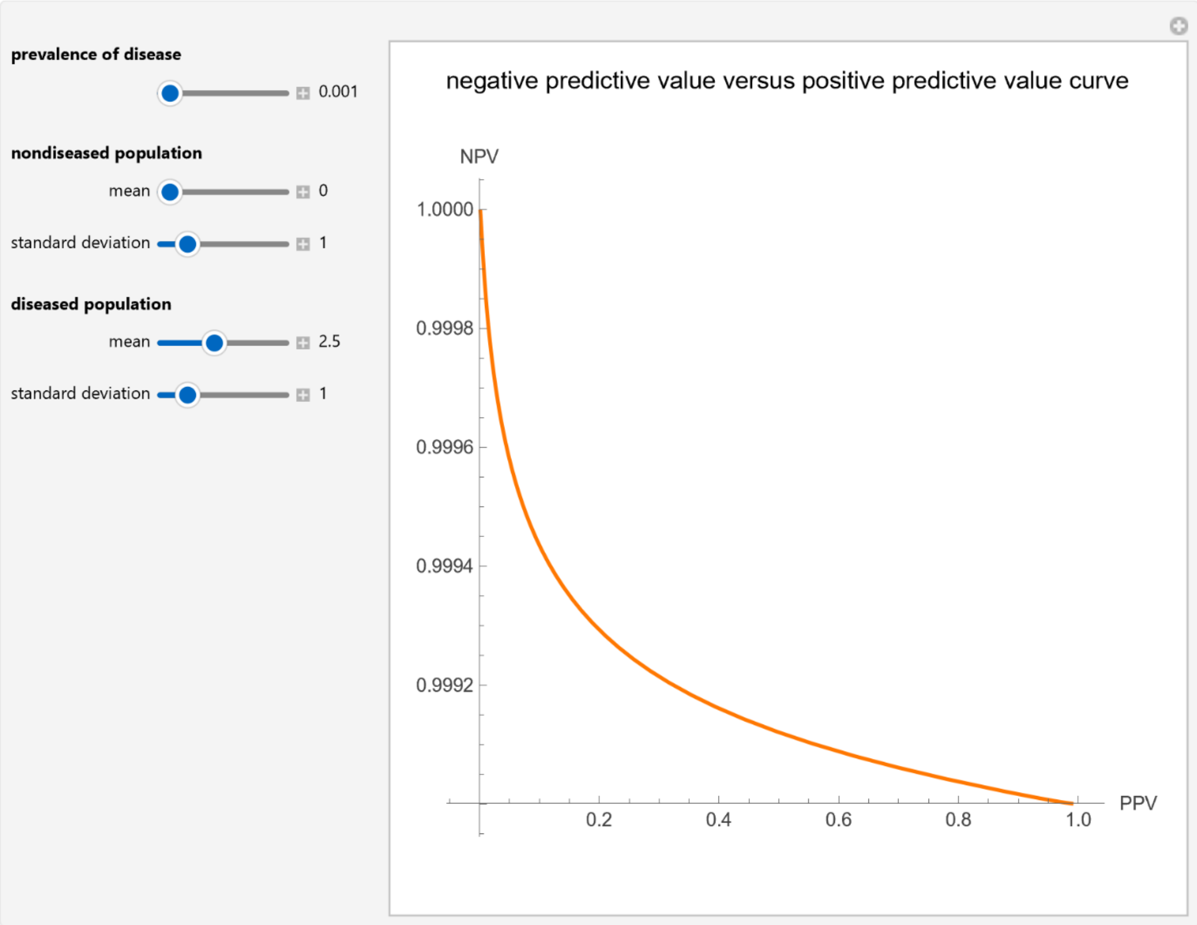

Figure 2: *Negative predictive value versus positive predictive value curve plot of a diagnostic test, with the settings shown on the left.*

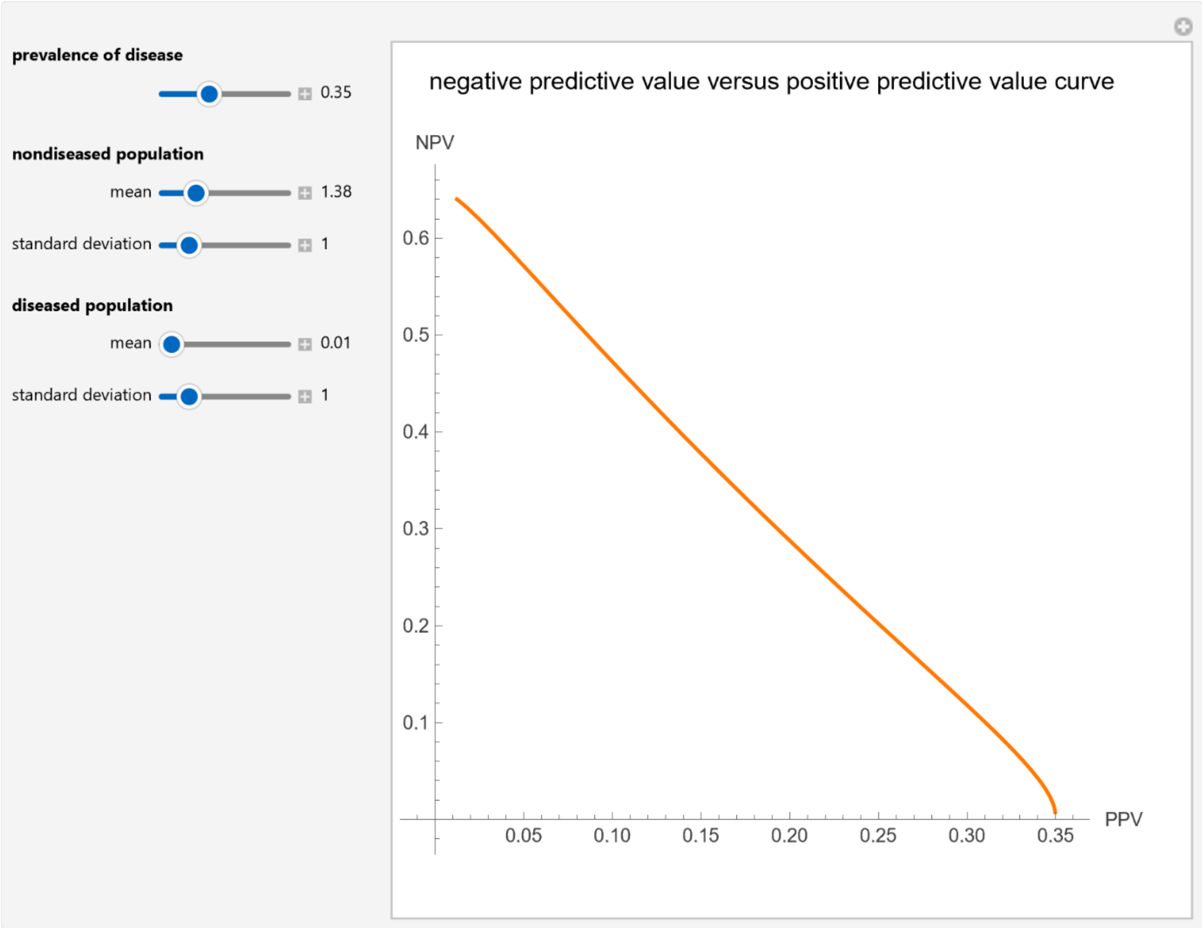

Figure 3: *Negative predictive value versus positive predictive value curve plot of a diagnostic test, with the settings shown on the left.*

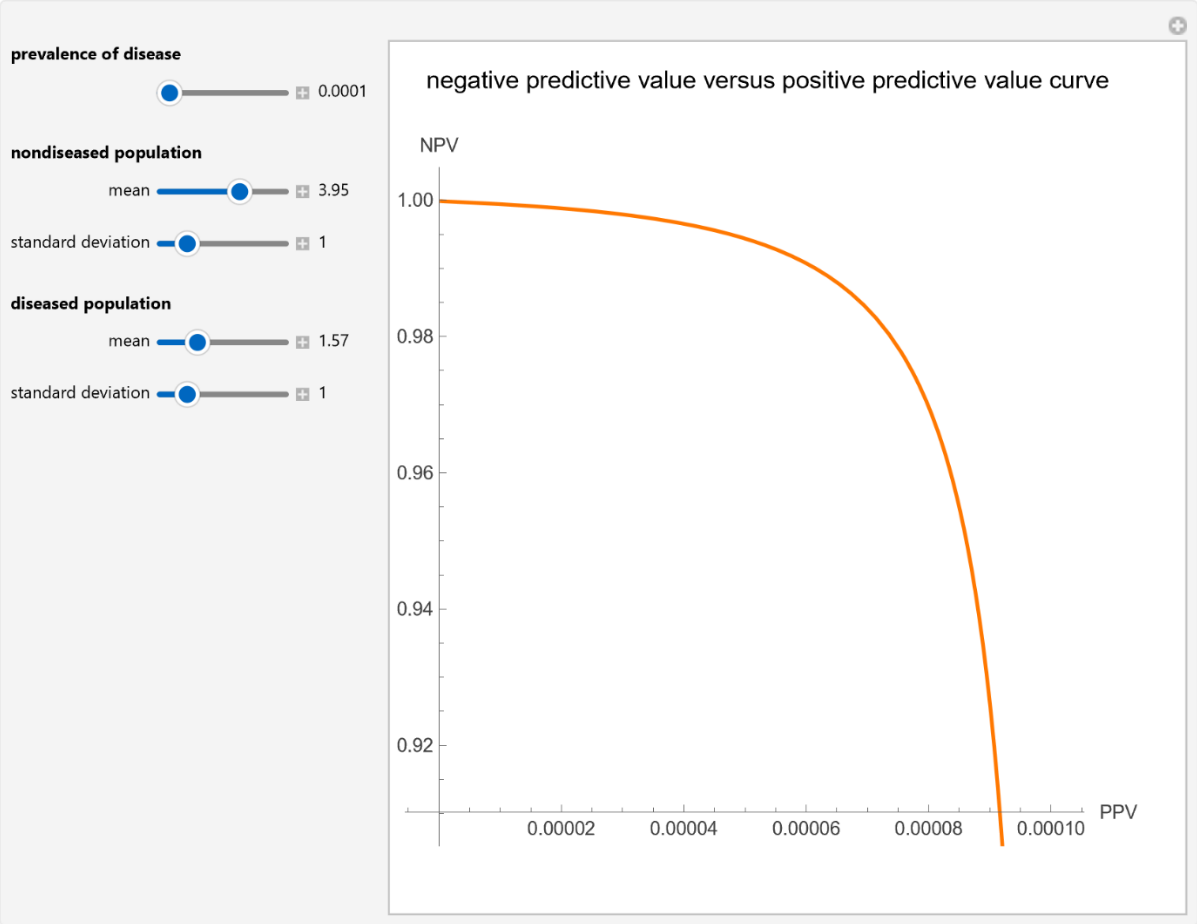

Figure 4: *Negative predictive value versus positive predictive value curve plot of a diagnostic test, with the settings shown on the left.*

Details

The PPV and NPV that are used in the evaluation of the clinical accuracy of a diagnostic test applied to a diseased and nondiseased population can be calculated given the sensitivity and specificity of the test. Sensitivity is the fraction of the diseased population with a positive test result, while specificity is the fraction of the nondiseased population with a negative test result. If we denote by Se the sensitivity, Sp the specificity, and v the prevalence, then:

$$PPV = \frac{Se\ v}{Se\ v + (1 - Sp)(1 - v)}$$

$$NPV = \frac{Sp\ (1 - v)}{Sp\ (1 - v) + (1 - v)\ pr}$$

Given a diseased and a nondiseased population, the specificity can be defined as a function of sensitivity; therefore, we can plot NPV versus PPV (a parametric plot).

The population data in the figures describe a bimodal distribution of serum glucose measurements in nondiabetic and diabetic populations [1].

This Demonstration is based on [2].

References

[1] T.-O. Lim, R. Bakri, Z. Morad and M. A. Hamid. Bimodality in Blood Glucose Distribution: Is It Universal? *Diabetes Care,* **25**(12), 2002 pp. 2212–2217.

[2] A. T. Hatjimihail, "Uncertainty of Measurement and Diagnostic Accuracy Measures", Ver. 1.1.2, Hellenic Complex Systems Laboratory, 2026. https://doi.org/10.5281/zenodo.20357353

Demonstration

DOI

10.5281/zenodo.20367402

Source

Programming language: Wolfram Language
Availability: The updated source notebook is available at:
https://www.hcsl.com/Tools/Demonstrations/RelationOfPositiveAndNegativePredictiveValuesOfDiagnostic.nb

First published in 2015 as part of the Wolfram Demonstrations Project:
https://demonstrations.wolfram.com/CorrelationOfPositiveAndNegativePredictiveValuesOfDiagnostic/

Software Requirements

Operating systems: Microsoft Windows, Linux, Apple macOS and iOS
Other software requirements: Wolfram Player®, freely available at: https://www.wolfram.com/player/ or Wolfram Mathematica®.

System Requirements

Processor: x86-64 compatible CPU.
System memory (RAM): 4 GB+ recommended.

Citation

Chatzimichail T. *Relation of Positive and Negative Predictive Values*. Ver. 1.1.1. Hellenic Complex Systems Laboratory; 2026. https://doi.org/10.5281/zenodo.20367402

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Technical Report

DOI

10.5281/zenodo.20367603

Citation

Chatzimichail T. *Relation of Positive and Negative Predictive Values*. Technical Report XII. Hellenic Complex Systems Laboratory; 2015. https://doi.org/10.5281/zenodo.20367603

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First Published: July 30, 2015

Revised: May 24, 2026